

2-1 Represent 4-Digit Numbers

Learning goal: I can show a 4-digit number with base-ten blocks and a place-value chart, and write it in standard form, expanded form, and word form.

Textbook: pages 33–36. The book models 2,138 with base-ten blocks, a place-value chart, and three written forms.

Words to Know

digit	One of the symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that we use to write numbers.
place value	The value a digit has because of its place. In 2,138 the 2 means 2 thousands, or 2,000.
base-ten blocks	A cube is 1,000. A flat is 100. A rod is 10. A unit is 1.
standard form	The usual way to write a number with digits: 2,138 .
expanded form	The number written as the value of each digit added together: 2,000 + 100 + 30 + 8 .
word form	The number written in words: two thousand, one hundred thirty-eight .

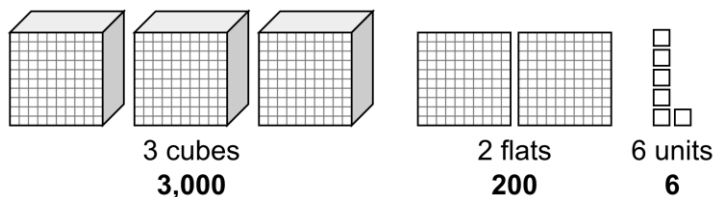
How to Teach It

1. Build or draw the number. Count the cubes, flats, rods, and units.
2. Write each count in a place-value chart: thousands, hundreds, tens, ones. If a place has none, write **0** there.
3. Read the chart from left to right to write **standard form**. Put a comma after the thousands digit.
4. For **expanded form**, write the value of each digit and join them with + signs. Skip places that have a 0 (6,000 + 400 + 9).
5. For **word form**, say the thousands, then the word “thousand,” then read the rest like a 3-digit number. Use a hyphen for numbers like forty-one.

Tip: Ask, “What is this digit worth?” A 0 holds a place. In 3,206 the 0 shows there are no tens.

Worked Examples

Example 1: Blocks to chart and standard form



thousands	hundreds	tens	ones
3	2	0	6

No rods, so write **0** in the tens place.

Standard form: **3,206**

Expanded form: **3,000 + 200 + 6**

Example 2: Standard form to expanded and word form

4,152 → the digits are worth 4,000, 100, 50, and 2.

Expanded form: **4,000 + 100 + 50 + 2**

Word form: **four thousand, one hundred fifty-two**

Example 3: Word or expanded form to standard form

“five thousand, seventy-three”

5 thousands, 0 hundreds, 7 tens, 3 ones → **5,073**

Expanded form: **5,000 + 70 + 3**

6,000 + 400 + 9 → 6 thousands, 4 hundreds, 0 tens, 9 ones → **6,409**

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|--|
| ☐ Tell what a cube, flat, rod, and unit are worth. | ☐ Write a number in expanded form. |
| ☐ Write the number shown by base-ten blocks. | ☐ Read and write a number in word form. |
| ☐ Fill in a place-value chart, using 0 for an empty place. | ☐ Use place value to explain how two numbers are different. |

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Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** What number do the base-ten blocks show? Fill in the chart. Then write the number in standard form.



thousands	hundreds	tens	ones

Standard form: _____

- B.** Write **3,081** in expanded form and in word form.

Expanded form: _____

Word form: _____

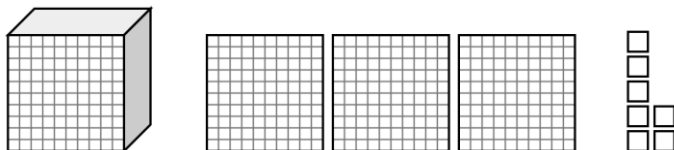
- C.** Write **four thousand, six hundred nine** in standard form and in expanded form.

Standard form: _____

Expanded form: _____

On My Own *Try these by yourself. Show your work.*

- 1.** What number do the base-ten blocks show? Fill in the chart.



thousands	hundreds	tens	ones

Standard form: _____

2. Show **5,264** in the place-value chart and in expanded form.

thousands	hundreds	tens	ones

Expanded form: _____

3. Write **eight thousand, fifty-six** in standard form and in expanded form.

Standard form: _____

Expanded form: _____

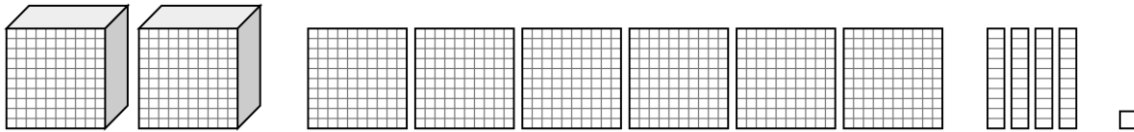
4. Write the number in standard form.

$$7,000 + 300 + 8$$

5. Write the number in standard form.

$$9,000 + 60 + 4$$

6. Write the number shown in standard form and in expanded form.



Standard form: _____

Expanded form: _____

7. A library had **2,374** books in May. In June it had **2,474** books. Which digit changed? Use place value to tell how many more books the library had in June.

8. Use the digits **6, 4, 9, 4** to write the number with the **least** value. Write it in standard form, expanded form, and word form.

Standard form: _____

Expanded form: _____

Word form: _____

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Together

A	Chart: 2 4 3 5. Standard form: 2,435 <i>2 cubes = 2,000; 4 flats = 400; 3 rods = 30; 5 units = 5.</i>
B	Expanded form: 3,000 + 80 + 1 Word form: three thousand, eighty-one <i>There are 0 hundreds, so no hundreds part is written. Also accept 3,000 + 0 + 80 + 1.</i>
C	Standard form: 4,609 Expanded form: 4,000 + 600 + 9 <i>There are 0 tens, so write 0 in the tens place. Also accept 4,000 + 600 + 0 + 9.</i>

On My Own

1	Chart: 1 3 0 7. Standard form: 1,307 <i>No rods, so the tens digit is 0. If your child writes 137, point to the empty tens place.</i>
2	Chart: 5 2 6 4 Expanded form: 5,000 + 200 + 60 + 4
3	Standard form: 8,056 Expanded form: 8,000 + 50 + 6 <i>0 hundreds. Also accept 8,000 + 0 + 50 + 6.</i>
4	7,308 <i>No tens, so the tens digit is 0.</i>
5	9,064 <i>No hundreds, so the hundreds digit is 0.</i>
6	Standard form: 2,641 Expanded form: 2,000 + 600 + 40 + 1
7	The hundreds digit changed from 3 to 4. That is 1 more hundred, so the library had 100 more books in June. <i>Accept any clear place-value explanation, e.g., "The thousands, tens, and ones are the same. Only the hundreds went up by one, so it is 100 more."</i>
8	Standard form: 4,469 Expanded form: 4,000 + 400 + 60 + 9 Word form: four thousand, four hundred sixty-nine <i>Put the smallest digits in the biggest places: 4, 4, 6, 9.</i>